

3 Superposition and Standing Waves

The wave model was introduced in the previous two chapters. We have seen that waves are very different from particles. A particle is of zero size, whereas a wave has a characteristic size, its wavelength. Another important difference between waves and particles is that we can explore the possibility of two or more waves combining at one point in the same medium. Particles can be combined to form extended objects, but the particles must be at different locations. In contrast, two waves can both be present at the same location. The ramifications of this possibility are explored in this chapter.

When waves are combined in systems with boundary conditions, only certain allowed frequencies can exist and we say the frequencies are quantized. Quantization is a notion that is at the heart of quantum mechanics. There we show that waves under boundary conditions explain many of the quantum phenomena. In this chapter, we use quantization to understand the behavior of the wide array of musical instruments that are based on strings and air columns. We also consider the combination of waves having different frequencies. When two sound waves having nearly the same frequency interfere, we hear variations in the loudness called beats. Finally, we discuss how any nonsinusoidal periodic wave can be described as a sum of sine and cosine functions.

3.1 Superposition and Interference

Many interesting wave phenomena in nature cannot be described by a single traveling wave. Instead, one must analyze these phenomena in terms of a combination of traveling waves. To analyze such wave combinations, we make use of the **superposition principle**:

> *If two or more traveling waves are moving through a medium, the resultant value of the wave function at any point is the algebraic sum of the values of the wave functions of the individual waves.*

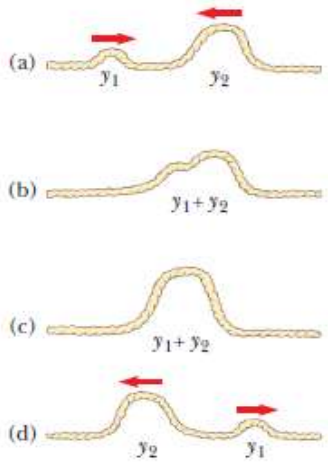
Waves that obey this principle are called linear waves. In the case of mechanical waves, linear waves are generally characterized by having amplitudes much smaller than their wavelengths. Waves that violate the superposition principle are called nonlinear waves and are often characterized by large amplitudes. In this book, we deal only with linear waves.

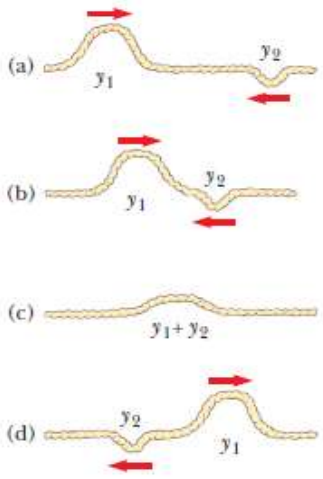
One consequence of the superposition principle is that two traveling waves can pass through each other without being destroyed or even altered. For instance, when two pebbles are thrown into a pond and hit the surface at different locations, the expanding circular surface waves from the two locations do not destroy each other but rather pass through each other. The resulting complex pattern can be viewed as two independent sets of expanding circles.

Active Figure 3.1 is a pictorial representation of the superposition of two pulses. The wave function for the pulse moving to the right is y_1, and the wave function for the pulse moving to the left is y_2. The pulses have the same speed but different shapes, and the displacement of the elements of the medium is in the positive y direction for both pulses. When the waves begin to overlap (Active Fig. 3.1b), the wave function for the resulting complex wave is given by $y_1 + y_2$. When the crests of the pulses coincide (Active Fig. 3.1c), the resulting wave given by $y_1 + y_2$ has a larger amplitude than that of the individual pulses. The two pulses finally separate and continue moving in their original directions (Active Fig. 3.1d). Notice that the pulse shapes remain unchanged after the interaction, as if the two pulses had never met!

The combination of separate waves in the same region of space to produce a resultant wave is called interference. For the two pulses shown in Active Figure 3.1, the displacement of the elements of the medium is in the positive y direction for both pulses, and the resultant pulse (created when the individual pulses overlap) exhibits an amplitude greater than that of either individual pulse. Because the displacements caused by the two pulses are in the same direction, we refer to their superposition as constructive interference.

Now consider two pulses traveling in opposite directions on a taut string where one pulse is inverted relative to the other as illustrated in Active Figure 3.2. When these pulses begin to overlap, the resultant pulse is given by $y_1 + y_2$, but the values of the function y_2 are negative. Again, the two pulses pass through each other; because the displacements caused by the two pulses are in opposite directions, however, we refer to their superposition as destructive interference. The superposition principle is the centerpiece of the waves in interference model. In many situations, both in acoustics and optics, waves combine according to this principle and exhibit interesting phenomena with practical applications





ACTIVE FIGURE 3.1 & 3.2

Superposition of Sinusoidal Waves

Let us now apply the principle of superposition to two sinusoidal waves traveling in the same direction in a linear medium. If the two waves are traveling to the right and have the same frequency, wavelength, and amplitude but differ in phase, we can express their individual wave functions as

$$y_1 = A \sin (kx - \omega t) \qquad y_2 = A \sin (kx - \omega t + \phi)$$

where, as usual, $k = 2\pi/\lambda, \omega = 2\pi f$ and f is the phase constant. Hence, the resultant wave function y is

$$y = y_1 + y_2 = A[\sin (kx - \omega t) + \sin (kx - \omega t + \phi)]$$

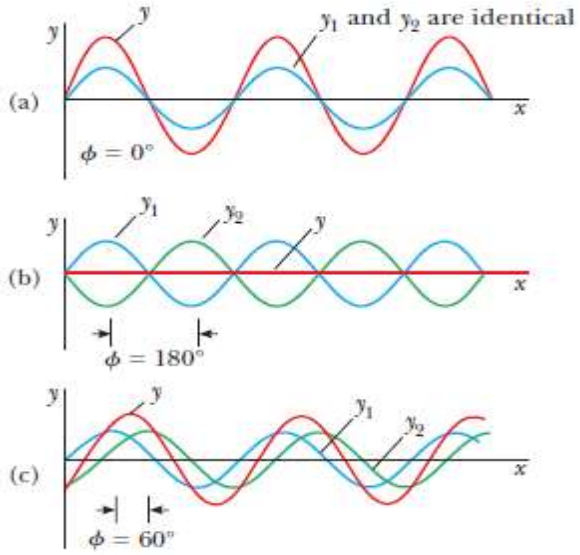
To simplify this expression, we use the trigonometric identity

$$\sin a + \sin b = 2 \cos \left(\frac{a - b}{2} \right) \sin \left(\frac{a + b}{2} \right)$$

Letting $a = kx - \omega t$ and $b = kx - \omega t + \phi$, we find that the resultant wave function y reduces to

$$y = 2A \cos \left(\frac{\phi}{2} \right) \sin \left(kx - \omega t + \frac{\phi}{2} \right)$$

This result has several important features. The resultant wave function y also is sinusoidal and has the same frequency and wavelength as the individual waves because the sine function incorporates the same values of k and v that appear in the original wave functions. The amplitude of the resultant wave is $2Acos(\emptyset/2)$, and its phase is $\emptyset/2$. If the phase constant $\emptyset$ equals 0, then cos ($\emptyset/2$) = cos 0 = 1 and the amplitude of the resultant wave is 2A, twice the amplitude of either individual wave. In this case, the waves are said to be everywhere in phase and therefore interfere constructively. That is, the crests and troughs of the individual waves y_1.



ACTIVE FIGURE 3.3

The superposition of two identical waves y_1 and y_2 (blue and green, respectively) to yield a resultant wave (red). (a) When y_1 and y_2 are in phase, the result is constructive interference. (b) When y_1 and y_2 are p rad out of phase, the result is destructive interference. (c) When the phase angle has a value other than 0 or π rad, the resultant wave *y* falls somewhere between the extremes shown in (a) and (b) and y_2 occur at the same positions and combine to form the red curve y of amplitude 2A shown in Active Figure 3.3a. Because the individual waves are in phase, they are indistinguishable in Active Figure 3.3a, in which they appear as a single blue curve. In general, constructive interference occurs when $\cos\left(\frac{\phi}{2}\right) = \pm 1$ That is true, for example, when ϕ = 0, 2π, 4π, . . . rad, that is, when f is an even multiple of π.

When ϕ is equal to π rad or to any odd multiple of π, then $\cos\left(\frac{\phi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$ and the crests of one wave occur at the same positions as the troughs of the second wave (Active Fig. 3.3b). Therefore, as a consequence of destructive interference, the resultant wave has zero amplitude everywhere. Finally, when the phase constant has an arbitrary value other than 0 or an integer multiple of π rad (Active Fig. 3.3c), the resultant wave has an amplitude whose value is somewhere between 0 and 2A.

In the more general case in which the waves have the same wavelength but different amplitudes, the results are similar with the following exceptions. In the in-phase case, the amplitude of the resultant wave is not twice that of a single wave, but rather is the sum of the amplitudes of the two waves. When the waves are π rad out of phase, they do not completely cancel as in Active Figure 3.3b. The result is a wave whose amplitude is the difference in the amplitudes of the individual waves.

Interference of Sound Waves

One simple device for demonstrating interference of sound waves is illustrated in Figure 3.4. Sound from a loudspeaker S is sent into a tube at point P, where there is a T-shaped junction. Half the sound energy travels in one direction, and half travels in the opposite direction. Therefore, the sound waves that reach the receiver R can travel along either of the two paths. The distance along any pat from speaker to receiver is called the **path length** r. The lower path length r_1 is fixed, but the upper path length r_2 can be varied by sliding the U-shaped tube, which is similar to that on a slide trombone. When the difference in the path lengths $\Delta r = |r_1 - r_2|$ is either zero or some integer multiple of the wavelength λ (that is, $\Delta r = n\lambda$, where n = 0, 1, 2, 3, . . .), the two waves reaching the receiver at any instant are in phase and interfere constructively as shown in Active Figure 3.3a. For this case, a maximum in the sound intensity is detected at the receiver. If the path length r_2 is adjusted such that the path difference $\Delta r = \frac{\lambda}{2}, \frac{3\lambda}{2}, \ldots, n\lambda/2$ (for n odd), the two waves are exactly π rad, or 180°, out of

phase at the receiver and hence cancel each other. In this case of destructive interference, no sound is detected at the receiver. This simple experiment demonstrates that a phase difference may arise between two waves generated by the same source when they travel along paths of unequal lengths.

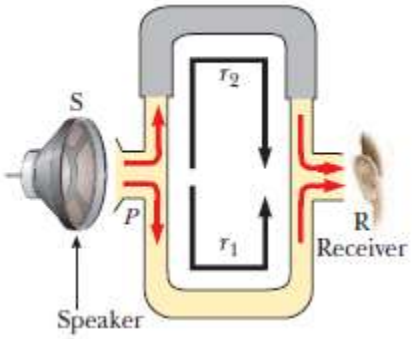


Figure 3.4: An acoustical system for demonstrating interference of sound waves.

3.2 Standing Waves

The sound waves from the pair of loudspeakers in Example 3.1 leave the speakers in the forward direction, and we considered interference at a point in front of the speakers. Suppose we turn the speakers so that they face each other and then have them emit sound of the same frequency and amplitude. In this situation, two identical waves travel in opposite directions in the same medium as in Figure 3.6. These waves combine in accordance with the waves in interference model.
We can analyze such a situation by considering wave functions for two transverse sinusoidal waves having the same amplitude, frequency, and wavelength but traveling in opposite directions in the same medium:

$$y_1 = A \sin (kx - \omega t) \qquad y_2 = A \sin (kx + \omega t)$$

where y_1 represents a wave traveling in the +x direction and y_2 represents one traveling in the -x direction. Adding these two functions gives the resultant wave function y:

$$y = y_1 + y_2 = A \sin (kx - \omega t) + A \sin (kx + \omega t)$$

When we use the trigonometric identity $\sin (a \pm b) = \sin (a) \cos (b) \pm \cos (a) \sin (b)$, this expression reduces to

$$y = (2A \sin kx) \cos \omega t \qquad [3.1]$$

Equation 3.1 represents the wave function of a standing wave. A standing wave such as the one on a string shown in Figure 3.7 is an oscillation pattern with a stationary outline that results from the superposition of two identical waves traveling in opposite directions.
Notice that Equation 3.1 does not contain a function of kx - ωt. Therefore, it is not an expression for a single traveling wave. When you observe a standing wave, there is no sense of motion in the direction of propagation of either original wave. Comparing Equation 3.1 with Equation 15.6, we see that it describes a special kind of simple harmonic motion. Every element of the medium oscillates in simple harmonic motion with the same angular frequency ω (according to the cos ωt factor in the equation). The amplitude of the simple harmonic motion of a given element (given by the factor 2A sin kx, the coefficient of the cosine function) depends on the location x of the element in the medium, however.

The amplitude of the simple harmonic motion of an element of the medium has a minimum value of zero when x satisfies the condition sin kx = 0, that is, when

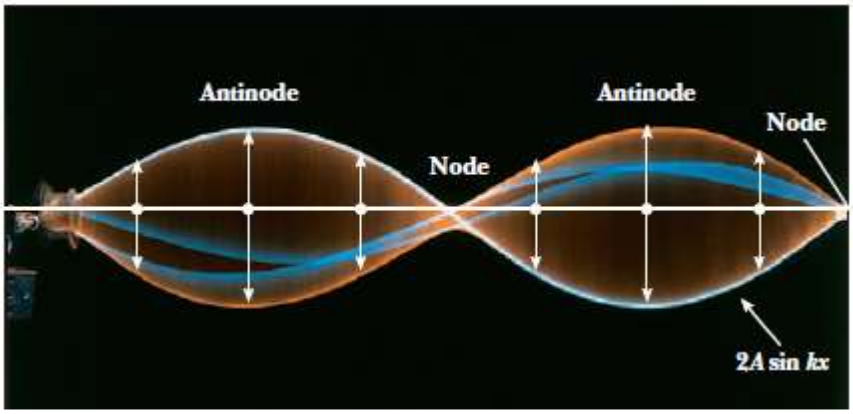


Figure 3.7 Multiflash photograph of a standing wave on a string

$$kx = 0, \pi, 2\pi, 3\pi, \ldots$$

Because $k = 2\pi/\lambda$, these values for kx give

$$x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \ldots = \frac{n\lambda}{2} \quad n = 0, 1, 2, 3, \ldots \qquad [3.2]$$

These points of zero amplitude are called nodes.

The element of the medium with the greatest possible displacement from equilibrium has an amplitude of 2A, which we define as the amplitude of the standing wave. The positions in the medium at which this maximum displacement occurs are called antinodes. The antinodes are located at positions for which the coordinate x satisfies the condition sin $kx = \pm 1$, that is, when

$$kx = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$$

Therefore, the positions of the antinodes are given by

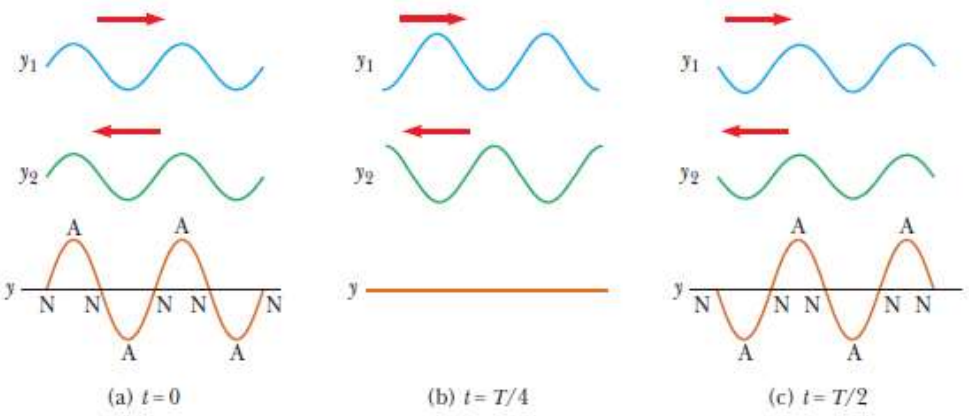
$$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots = \frac{n\lambda}{4} \quad n = 1, 3, 5, \ldots \qquad [3.3]$$

Two nodes and two antinodes are labeled in the standing wave in Figure 3.7. The light blue curve labeled 2A sin kx in Figure 3.7 represents one wavelength of the traveling waves that combine to form the standing wave. Figure 3.7 and Equations 3.2 and 3.3 provide the following important features of the locations of nodes and antinodes:

The distance between adjacent antinodes is equal to $\lambda/2$.
The distance between adjacent nodes is equal to $\lambda/2$.
The distance between a node and an adjacent antinode is $\lambda/4$.

Wave patterns of the elements of the medium produced at various times by two transverse traveling waves moving in opposite directions are shown in Active Figure 3.8. The blue and green curves are the wave patterns for the individual traveling waves, and the brown curves are the wave patterns for the resultant standing wave. At t = 0 (Active Fig. 3.8a), the two traveling waves are in phase, giving a

wave pattern in which each element of the medium is at rest and experiencing its maximum displacement from equilibrium.



ACTIVE FIGURE 3.8

One quarter of a period later, at t=T/4 (Active Fig. 3.8b), the traveling waves have moved one quarter of a wavelength (one to the right and the other to the left). At this time, the traveling waves are out of phase, and each element of the medium is passing through the equilibrium position in its simple harmonic motion. The result is zero displacement for elements at all values of x; that is, the wave pattern is a straight line. At t = T/2 (Active Fig. 3.8c), the traveling waves are again in phase, producing a wave pattern that is inverted relative to the t = 0 pattern. In the standing wave, the elements of the medium alternate in time between the extremes shown in Active Figure 3.8a and c.

3.3 Standing Waves in a String Fixed at Both Ends

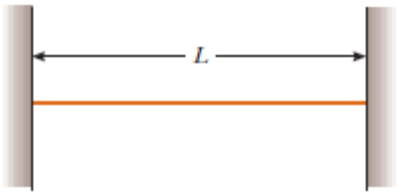
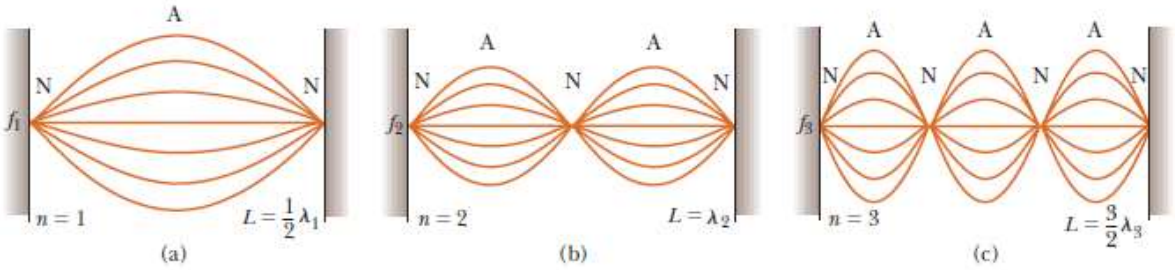


Figure 3.9 A string of length L fixed at both ends.

Consider a string of length L fixed at both ends as shown in Figure 3.9. We will use this system as a model for a guitar string or piano string. Standing waves can be set up in the string by a continuous superposition of waves incident on and reflected from the ends. Notice that there is a boundary condition for the waves on the string. Because the ends of the string are fixed, they must necessarily have zero displacement and are therefore nodes by definition. This boundary condition results in the string having a number of discrete natural patterns of oscillation, called normal modes, each of which has a characteristic frequency that is easily calculated. This situation in which only certain frequencies of oscillation are allowed is called quantization. Quantization is a common occurrence when waves are subject to boundary conditions and is a central feature in our discussions of quantum physics in the extended version of this text. Notice in Active Figure 3.8 that there are no boundary conditions, so standing waves of any frequency can be established; there is no quantization without boundary conditions. Because boundary conditions occur so often for waves, we identify an analysis model called the waves under boundary conditions model for the discussion that follows.

The normal modes of oscillation for the string in Figure 3.9 can be described by imposing the boundary conditions that the ends be nodes and that the nodes and antinodes be separated by one-

fourth of a wavelength. The first normal mode that is consistent with these requirements, shown in Active Figure 3.10a, has nodes at its ends and one antinode in the middle. This normal mode is the longest-wavelength mode that is consistent with our boundary conditions. The first normal mode occurs when the wavelength λ_1 is equal to twice the length of the string, or $\lambda_1 = 2L$.



ACTIVE FIGURE 3.10

The section of a standing wave from one node to the next node is called a loop. In the first normal mode, the string is vibrating in one loop. In the second normal mode (see Active Fig. 3.10b), the string vibrates in two loops. In this case, the wavelength λ_2 is equal to the length of the string, as expressed by $\lambda_2 = L$. The third normal mode (see Active Fig. 3.10c) corresponds to the case in which $\lambda_3 = 2L/3$, and our string vibrates in three loops. In general, the wavelengths of the various normal modes for a string of length L fixed at both ends are

$$\lambda_n = \frac{2L}{n} \quad n = 1, 2, 3, \ldots \qquad [3.4]$$

where the index n refers to the nth normal mode of oscillation. These nodes are the possible modes of oscillation for the string. The actual modes that are excited on a string are discussed shortly.

The natural frequencies associated with the modes of oscillation are obtained from the relationship $f = v/\lambda$, where the wave speed v is the same for all frequencies. Using Equation 3.4, we find that the natural frequencies fn of the normal modes are

$$f_n = \frac{v}{\lambda_n} = n\frac{v}{2L} \quad n = 1, 2, 3, \ldots \qquad [3.5]$$

These natural frequencies are also called the quantized frequencies associated with the vibrating string fixed at both ends. Because $v = \sqrt{\frac{T}{\mu}}$ for waves on a string, where T is the tension in the string and μ is its linear mass density, we can also express the natural frequencies of a taut string as

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}} \quad n = 1, 2, 3, \ldots \qquad [3.6]$$

The lowest frequency f_1, which corresponds to n = 1, is called either the fundamental or the fundamental frequency and is given by

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}} \qquad [3.7]$$

The frequencies of the remaining normal modes are integer multiples of the fundamental frequency. Frequencies of normal modes that exhibit an integer-multiple relationship such as this form a harmonic series, and the normal modes are called harmonics. The fundamental frequency f_1 is the frequency of the first harmonic, the frequency $f_2 = 2f_1$ is the frequency of the second harmonic, and the frequency $f_n = nf_1$ is the frequency of the nth harmonic. Other oscillating systems, such as a drumhead, exhibit normal modes, but the frequencies are not related as integer multiples of a fundamental (see Section 3.6). Therefore, we do not use the term harmonic in association with these types of systems. Let us examine further how the various harmonics are created in a string. To excite only a single harmonic, the string must be distorted into a shape that corresponds to that of the desired harmonic. After being released, the string vibrates at the frequency of that harmonic. This maneuver is difficult to perform, however, and is not how a string of a musical instrument is excited. If the string is distorted such that its shape is not that of just one harmonic, the resulting vibration
includes a combination of various harmonics. Such a distortion occurs in musical instruments when the string is plucked (as in a guitar), bowed (as in a cello), or struck (as in a piano). When the string is distorted into a nonsinusoidal shape, only waves that satisfy the boundary conditions can persist on the string. These waves are the harmonics. The frequency of a string that defines the musical note that it plays is that of the fundamental. The string's frequency can be varied by changing either the string's tension or its length. For example, the tension in guitar and violin strings is varied by a screw adjustment mechanism or by tuning pegs located on the neck of the instrument. As the tension is increased, the frequency of the normal modes increases in accordance with Equation 3.6. Once the instrument is "tuned," players vary the frequency by moving their fingers along the neck, thereby changing the length of the oscillating portion of the string. As the length is shortened, the frequency increases because, as Equation 3.6 specifies, the normal-mode frequencies are inversely proportional to string length.

3.4 Resonance

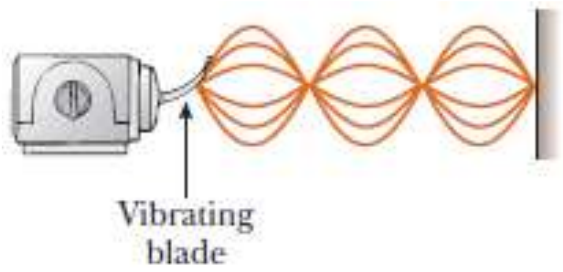


Figure 3.12 Standing waves are set up in a string when one end is connected to a vibrating blade. When the blade vibrates at one of the natural frequencies of the string, large amplitude standing waves are created.

We have seen that a system such as a taut string is capable of oscillating in one or more normal modes of oscillation. If a periodic force is applied to such a system, the amplitude of the resulting motion is greatest when the frequency of the applied force is equal to one of the natural frequencies of the system. This phenomenon, known as resonance, was discussed in Section 15.7. Although a block–spring system or a simple pendulum has only one natural frequency, standing-wave systems have a whole set of natural frequencies, such as that given by Equation 3.6 for a string. Because an oscillating

system exhibits a large amplitude when driven at any of its natural frequencies, these frequencies are often referred to as **resonance frequencies**.

Consider a taut string fixed at one end and connected at the opposite end to an oscillating blade as illustrated in Figure 3.12. The fixed end is a node, and the end connected to the blade is very nearly a node because the amplitude of the blade's motion is small compared with that of the elements of the string. As the blade oscillates, transverse waves sent down the string are reflected from the fixed end. As we learned in Section 3.3, the string has natural frequencies that are determined by its length, tension, and linear mass density (see Eq. 3.6). When the frequency of the blade equals one of the natural frequencies of the string, standing waves are produced and the string oscillates with a large amplitude. In this resonance case, the wave generated by the oscillating blade is in phase with the reflected wave and the string absorbs energy from the blade. If the string is driven at a frequency that is not one of its natural frequencies, the oscillations are of low
amplitude and exhibit no stable pattern. Resonance is very important in the excitation of musical instruments based on air columns. We shall discuss this application of resonance in Section 3.5.

3.5 Standing Waves in Air Columns

The waves under boundary conditions model can also be applied to sound waves in a column of air such as that inside an organ pipe. Standing waves are the result of interference between longitudinal sound waves traveling in opposite directions. In a pipe closed at one end, the closed end is a displacement node because the rigid barrier at this end does not allow longitudinal motion of the air. Because the pressure wave is 90° out of phase with the displacement wave the closed end of an air column corresponds to a pressure antinode (that is, a point of maximum pressure variation).

The open end of an air column is approximately a displacement antinode1 and a pressure node. We can understand why no pressure variation occurs at an open end by noting that the end of the air column is open to the atmosphere; therefore, the pressure at this end must remain constant at atmospheric pressure. You may wonder how a sound wave can reflect from an open end because there may not appear to be a change in the medium at this point: the medium through which the sound wave moves is air both inside and outside the pipe. Sound is a pressure wave, however, and a compression region of the sound wave is constrained by the sides of the pipe as long as the region is inside the pipe. As the compression region exits at the open end of the pipe, the constraint of the pipe is removed and the compressed air is free to expand into the atmosphere. Therefore, there is a change in the character of the medium between the inside of the pipe and the outside even though there is no change in the material of the medium. This change in character is sufficient to allow some reflection.

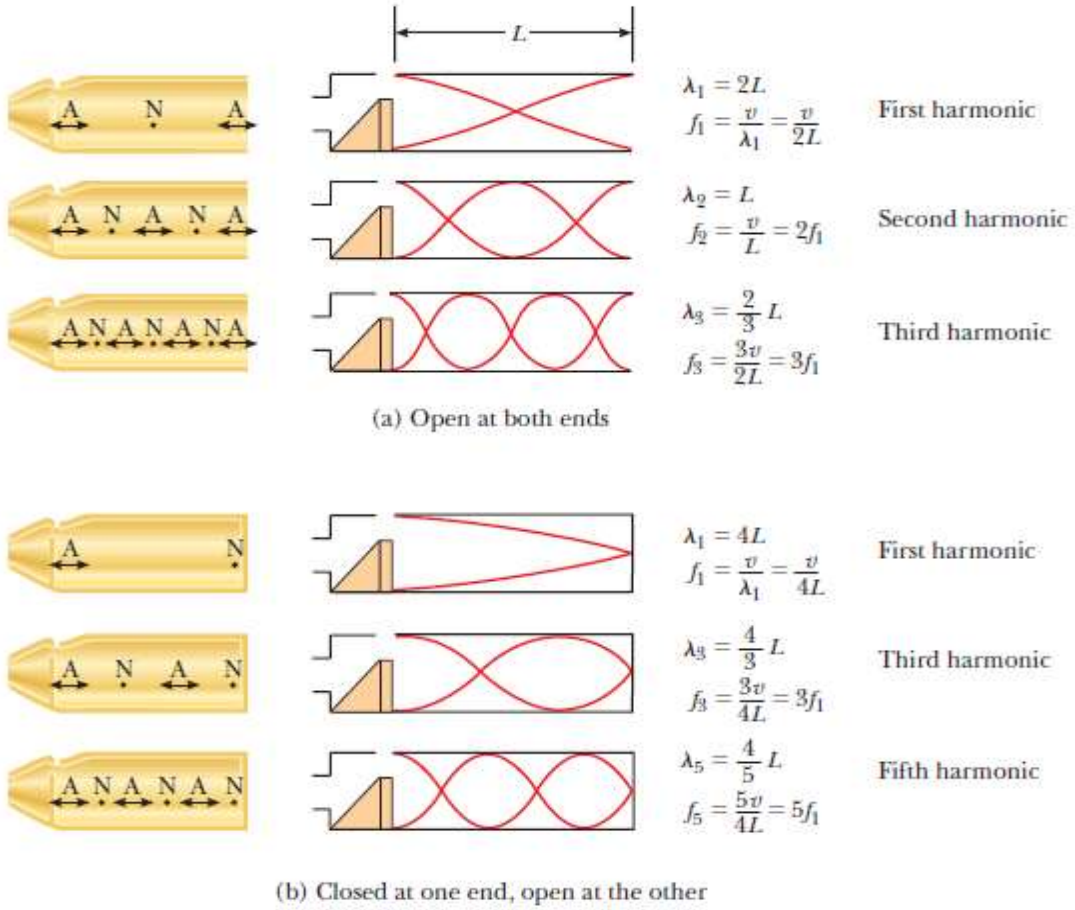


Figure 3.13 Motion of elements of air in standing longitudinal waves in a pipe, along with schematic representations of the waves.

With the boundary conditions of nodes or antinodes at the ends of the air column, we have a set of normal modes of oscillation as is the case for the string fixed at both ends. Therefore, the air column has quantized frequencies.

The first three normal modes of oscillation of a pipe open at both ends are shown in Figure 3.13a. Notice that both ends are displacement antinodes (approximately). In the first normal mode, the standing wave extends between two adjacent antinodes, which is a distance of half a wavelength. Therefore, the wavelength is twice the length of the pipe, and the fundamental frequency is $f_1 = v/2L$. As Figure 18.13a shows, the frequencies of the higher harmonics are $2f_1$, $3f_1$, ...

> *In a pipe open at both ends, the natural frequencies of oscillation form a harmonic series that includes all integral multiples of the fundamental frequency*.

Because all harmonics are present and because the fundamental frequency is given by the same expression as that for a string (see Eq. 3.5), we can express the natural frequencies of oscillation as

$$f_n = n\frac{v}{2L} \quad n = 1, 2, 3, \ldots \qquad [3.8]$$

Despite the similarity between Equations 3.5 and 3.8, you must remember that v in Equation 3.5 is the speed of waves on the string, whereas v in Equation 3.8 is the speed of sound in air.

If a pipe is closed at one end and open at the other, the closed end is a displacement node (see Fig. 3.13b). In this case, the standing wave for the fundamental mode extends from an antinode to the adjacent node, which is one-fourth of a wavelength. Hence, the wavelength for the first normal mode is 4L, and the fundamental frequency is $f_1 = v/4L$. As Figure 3.13b shows, the higher-frequency waves

that satisfy our conditions are those that have a node at the closed end and an antinode at the open end; hence, the higher harmonics have frequencies $3f_1$, $5f_1$, ...

> *In a pipe closed at one end, the natural frequencies of oscillation form a harmonic series that includes only odd integral multiples of the fundamental frequency.*

We express this result mathematically as

$$f_n = n\frac{v}{4L} \quad n = 1, 3, 5, \ldots \tag{3.9}$$

It is interesting to investigate what happens to the frequencies of instruments based on air columns and strings during a concert as the temperature rises. The sound emitted by a flute, for example, becomes sharp (increases in frequency) as the flute warms up because the speed of sound increases in the increasingly warmer air inside the flute (consider Eq. 3.8). The sound produced by a violin becomes flat (decreases in frequency) as the strings thermally expand because the expansion causes their tension to decrease (see Eq. 3.6). Musical instruments based on air columns are generally excited by resonance. The air column is presented with a sound wave that is rich in many frequencies. The air column then responds with a large-amplitude oscillation to the frequencies that match the quantized frequencies in its set of harmonics. In many woodwind instruments, the initial rich sound is provided by a vibrating reed. In brass instruments, this excitation is provided by the sound coming from the vibration of the player's lips. In a flute, the initial excitation comes from blowing over an edge at the mouthpiece of the instrument in a manner similar to blowing across the
opening of a bottle with a narrow neck. The sound of the air rushing across the edge has many frequencies, including one that sets the air cavity in the bottle into resonance.

3.6 Standing Waves in Rods and Membranes

Standing waves can also be set up in rods and membranes. A rod clamped in the middle and stroked parallel to the rod at one end oscillates as depicted in Figure 3.15a. The oscillations of the elements of the rod are longitudinal, and so the red curves in Figure 3.15 represent longitudinal displacements of various parts of the rod. For clarity, the displacements have been drawn in the transverse direction as they were for air columns. The midpoint is a displacement node because it is fixed by the clamp, whereas the ends are displacement antinodes because they are free to oscillate. The oscillations in this setup are analogous to those in a pipe open at both ends. The red lines in Figure 3.15a represent the first normal mode, for which the wavelength is 2L and the frequency is f = v/2L, where v is the speed of longitudinal waves in the rod. Other normal modes may be excited by clamping the rod at different points. For example, the second normal mode (Fig. 3.15b) is excited by clamping the rod a distance L/4 away from one end.

It is also possible to set up transverse standing waves in rods. Musical instruments that depend on transverse standing waves in rods include triangles, marimbas,
xylophones, glockenspiels, chimes, and vibraphones. Other devices that make sounds from vibrating bars include music boxes and wind chimes.

Two-dimensional oscillations can be set up in a flexible membrane stretched over a circular hoop such as that in a drumhead. As the membrane is struck at
some point, waves that arrive at the fixed boundary are reflected many times. The resulting sound is not harmonic because the standing waves have frequencies that are not related by integer multiples. Without this relationship, the sound may be more correctly described as noise rather than as music.

The production of noise is in contrast to the situation in wind and stringed instruments, which produce sounds that we describe as musical.

Some possible normal modes of oscillation for a two-dimensional circular membrane are shown in Figure 3.16. Whereas nodes are points in one-dimensional standing waves on strings and in air columns, a two-dimensional oscillator has curves along which there is no displacement of the elements of the medium. The lowest normal mode, which has a frequency f_1, contains only one nodal curve; this curve runs around the outer edge of the membrane. The other possible normal modes show additional nodal curves that are circles and straight lines across the diameter of the membrane.

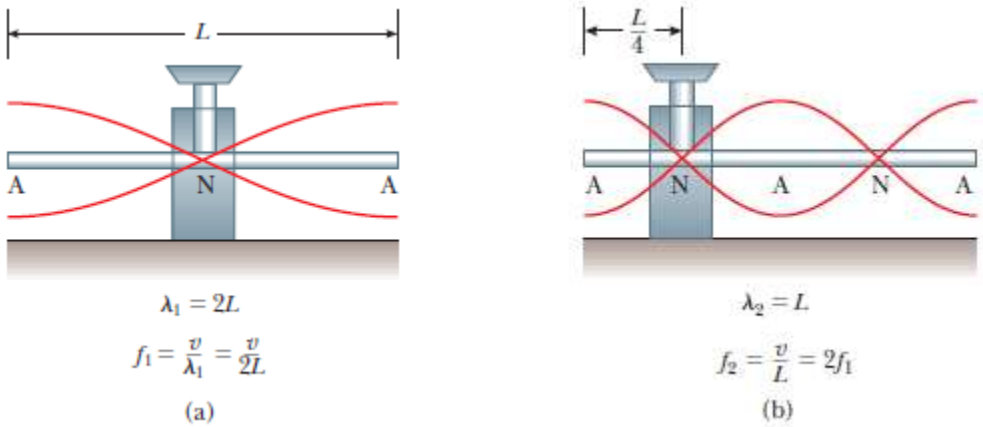


Figure 3.15 Normal-mode longitudinal vibrations of a rod of length L

PROBLEMS

1. Two waves in one string are described by the wave functions

$$y_1 = (3.0\ \text{cm}) \cos (4.0x - 1.6t)$$
$$y_2 = (4.0\ \text{cm}) \sin (5.0x - 2.0t)$$

   where y and x are in centimeters and t is in seconds. Find the superposition of the waves y1 _ y2 at the points (a) x = 1.00, t = 1.00; (b) x = 1.00, t = 0.500; and (c) x = 0.500, t = 0. (Remember that the arguments of the trigonometric functions are in radians.)
2. Two pulses A and B are moving in opposite directions along a taut string with a speed of 2.00 cm/s. The amplitude of A is twice the amplitude of B. The pulses are shown in Figure P3.2 at t= 0. Sketch the shape of the string at t = 1, 1.5, 2, 2.5, and 3 s.

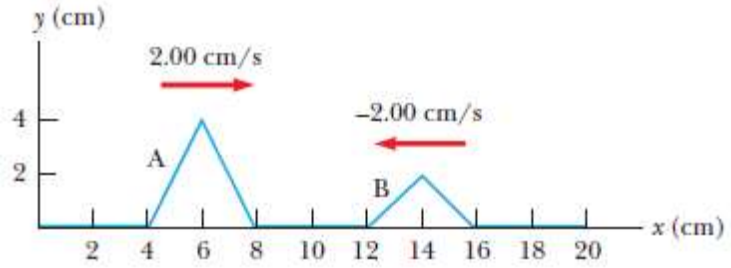


Figure P3.2

3. Two pulses traveling on the same string are described by

$$y_1 = \frac{5}{(3x - 4t)^2 + 2} \qquad y_2 = \frac{-5}{(3x + 4t - 6)^2 + 2}$$

(a) In which direction does each pulse travel? (b) At what instant do the two cancel everywhere? (c) At what point do the two pulses always cancel?

4. Two waves are traveling in the same direction along a stretched string. The waves are 90.0° out of phase. Each wave has an amplitude of 4.00 cm. Find the amplitude of the resultant wave.
5. Two traveling sinusoidal waves are described by the wave functions

$$y_1 = (5.00\ \text{m}) \sin [\pi(4.00x - 1\,200t)]$$

$$y_2 = (5.00\ \text{m}) \sin [\pi(4.00x - 1\,200t - 0.250)]$$

where x, y_1, and y_2 are in meters and t is in seconds. (a) What is the amplitude of the resultant wave? (b) What is the frequency of the resultant wave?

6. Two identical loudspeakers are placed on a wall 2.00 m apart. A listener stands 3.00 m from the wall directly in front of one of the speakers. A single oscillator is driving the speakers at a frequency of 300 Hz. (a) What is the phase difference between the two waves when they reach the observer? (b) **What If**? What is the frequency closest to 300 Hz to which the oscillator may be adjusted such that the observer hears minimal sound?
7. Two identical loudspeakers are driven by the same oscillator of frequency 200 Hz. The speakers are located on a vertical pole a distance of 4.00 m from each other. A man walks straight toward the lower speaker in a direction perpendicular to the pole as shown in Figure P3.7. (a) How many times will he hear a minimum in sound intensity? (b) How far is he from the pole at these moments? Take the speed of sound to be 330 m/s and ignore any sound reflection from the ground.

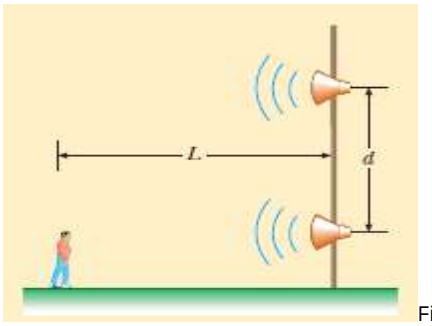


Figure P3.7 Problems 7 and 8.

8. Two identical loudspeakers are driven by the same oscillator of frequency f. The speakers are located a distance d from each other on a vertical pole. A man walks straight toward the lower speaker in a direction perpendicular to the pole as shown in Figure P3.7. (a) How many times will he hear a minimum in sound intensity? (b) How far is he from the pole at these moments? Let v represent the speed of sound and assume the ground does not reflect sound.
9. Two sinusoidal waves in a string are defined by the functions

$$y_1 = (2.00\ \text{cm}) \sin (20.0x - 32.0t)$$

$$y_2 = (2.00\ \text{cm}) \sin (25.0x - 40.0t)$$

where y_1, y_2, and x are in centimeters and t is in seconds. (a) What is the phase difference between these two waves at the point x = 5.00 cm at t = 2.00 s? (b) What is the positive x value

closest to the origin for which the two phases differ by $\pm\pi$ at t = 2.00 s? (That is a location where the two waves add to zero.)

10. Find the fundamental frequency and the next three frequencies that could cause standing wave patterns on a string that is 30.0 m long, has a mass per unit length of 9.00×10^{-3} kg/m, and is stretched to a tension of 20.0 N.
11. A string with a mass of 8.00 g and a length of 5.00 m has one end attached to a wall; the other end is draped over a small fixed pulley and attached to a hanging object with a mass of 4.00 kg. If the string is plucked, what is the fundamental frequency of its vibration?
12. In the arrangement shown in Figure P3.8, an object can be hung from a string (with linear mass density μ = 0.002 00 kg/m) that passes over a light pulley. The string is connected to a vibrator (of constant frequency f), and the length of the string between point P and the pulley is L = 2.00 m. When the mass m of the object is either 16.0 kg or 25.0 kg, standing waves are observed; no standing waves are observed with any mass between these values, however. (a) What is the frequency of the vibrator? Note: The greater the tension in the string, the smaller the number of nodes in the standing wave. (b) What is the largest object mass for which standing waves could be observed?

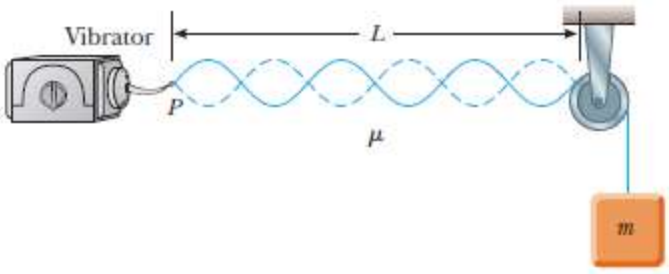


Figure P3.8